

DATA VISUALIZATION PORTFOLIO

About Me 1

Portfolio 1

Scientific publications 2

Infographics 6

Live Dashboards 9

About Me

I am a data scientist and researcher with a background in biomedical science, evidence synthesis and real-world clinical data. My work focuses on transforming complex datasets into clear, informative visualizations that support scientific communication and decision making. I primarily work in R using ggplot2, Quarto and Shiny, and enjoy designing visualizations that are both accurate and intuitive.

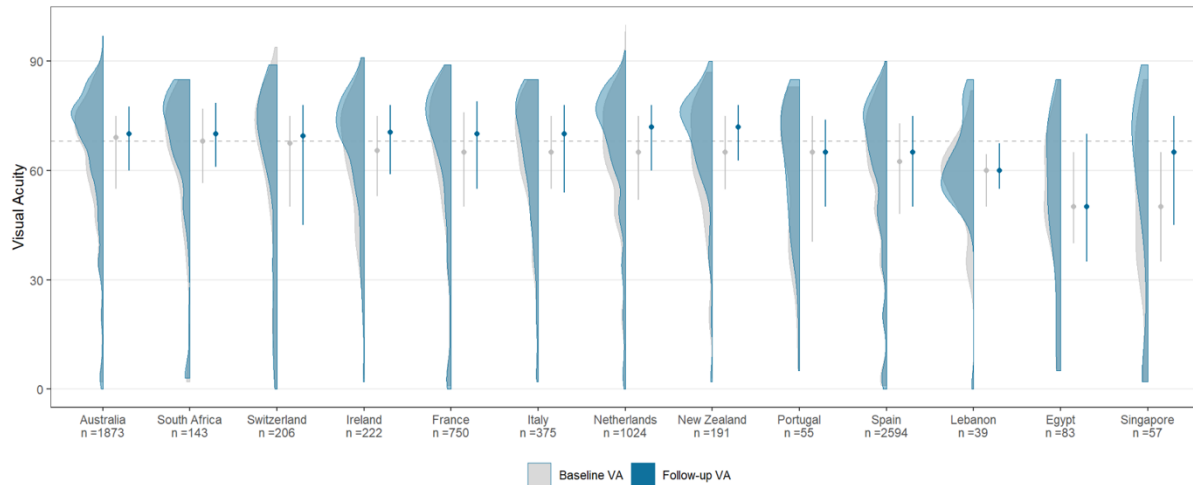
Portfolio

I enjoy using data visualization to communicate science. Throughout my work with systematic reviews, clinical registries and real-world evidence, I have developed visualizations that support data exploration, scientific publications and decision making. This portfolio presents a selection of static figures, infographics and interactive dashboards created for researchers, clinicians and other scientific audiences.

Scientific publications

International Comparison of Outcomes in Ophthalmology

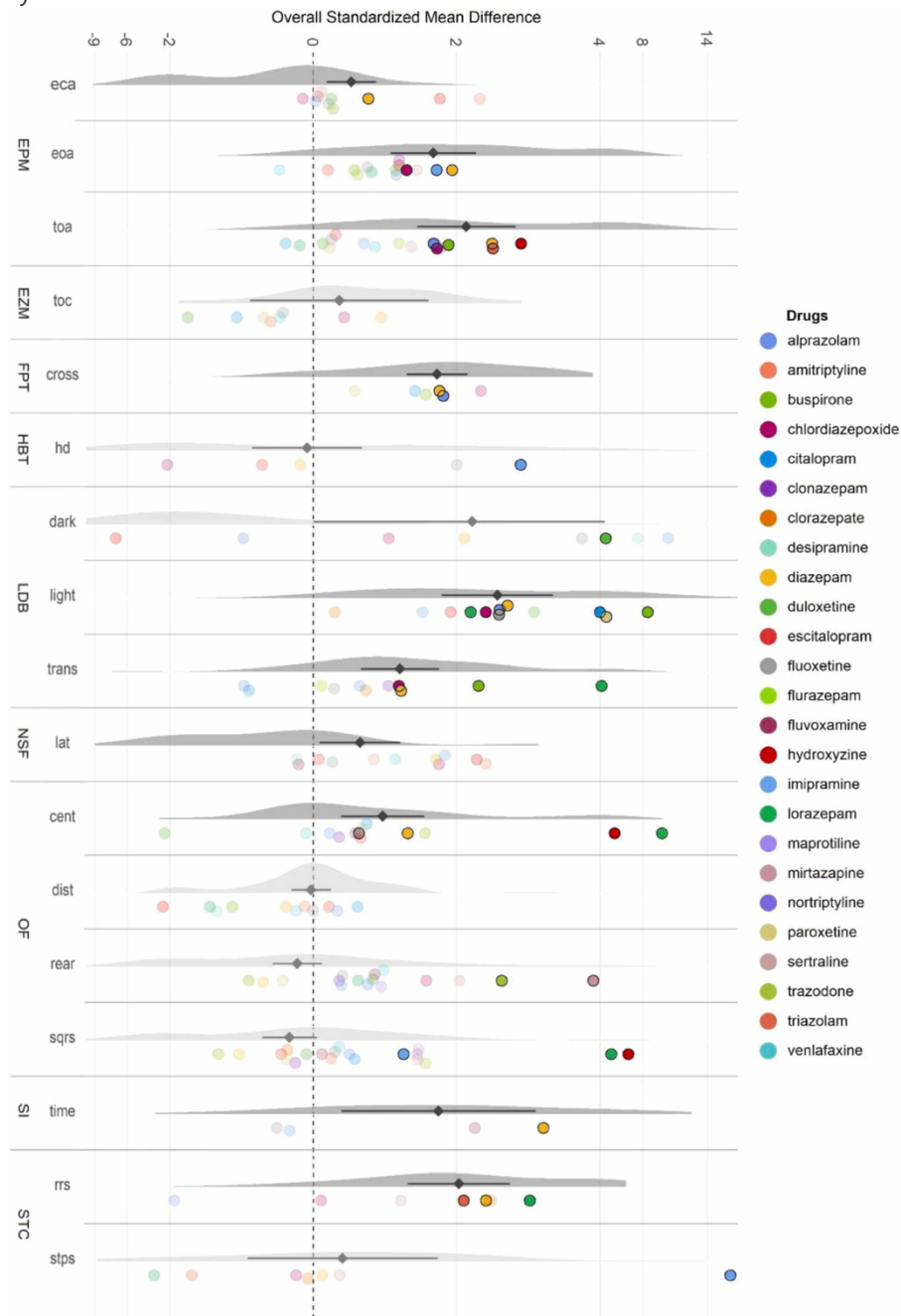
This project investigated international differences in the management of neovascular age-related macular degeneration (nAMD) using data from an international real-world ophthalmology registry. By comparing treatment practices and patient outcomes across countries with different healthcare systems, regulations, and access to care, the study aimed to identify factors associated with improved outcomes and inform best practices for nAMD management (Unpublished data - Publication in print)



Comparison of Baseline and Follow-up Visual Acuity (VA) Across Countries: The figure compares baseline and follow-up visual acuity (VA) across countries. Violin plots show VA distribution, with dashed lines indicating a reference value. Baseline VA is muted, follow-up VA is colored. Median values (points) and interquartile ranges (lines) are displayed. The x-axis orders countries by median VA values, and the y-axis represents VA measurements, offering insights into inter-country differences and trends.

Reliability of Anxiolytic Compounds

A large-scale evidence synthesis investigating the validity of preclinical mouse anxiety models by integrating data from 814 studies evaluating 25 clinically approved anxiolytics.



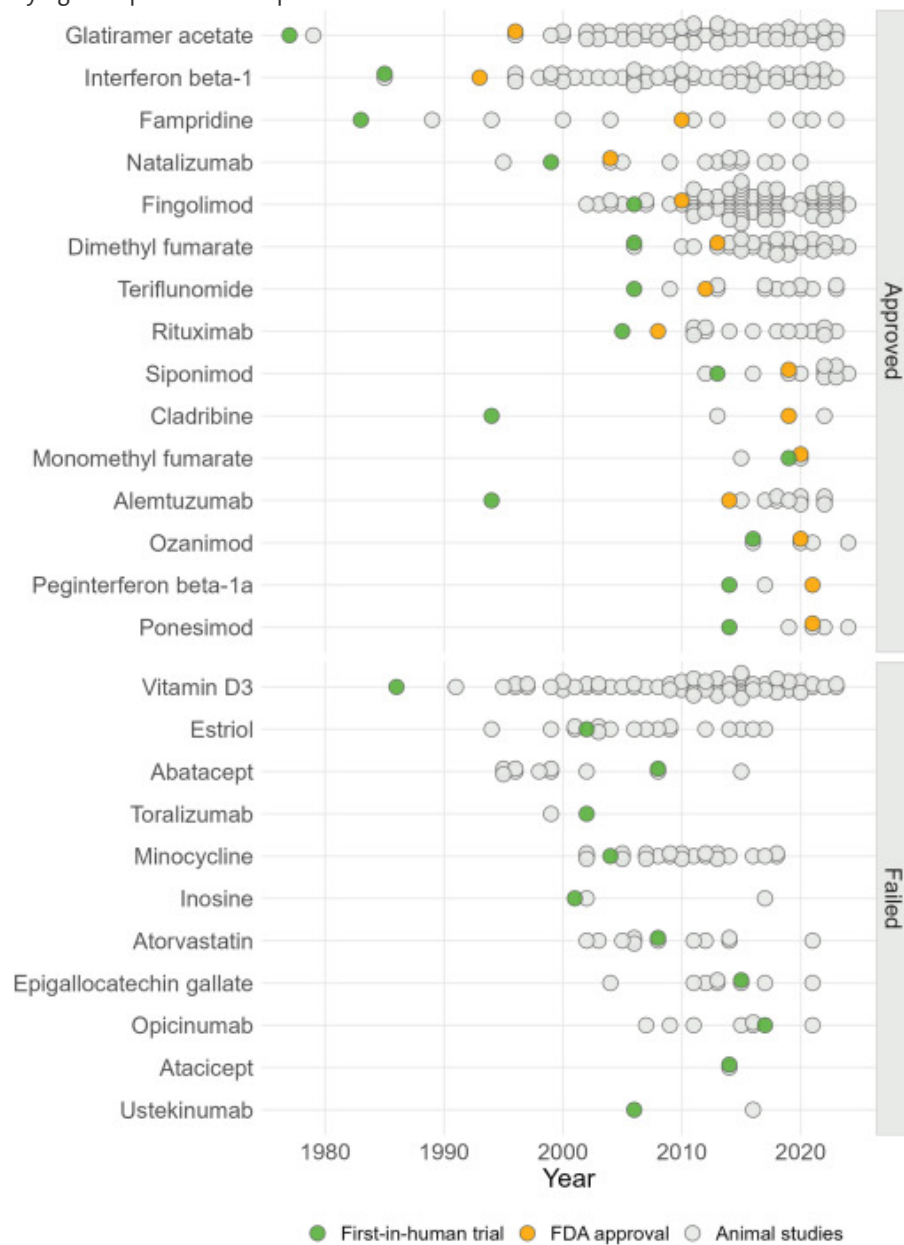
Distribution plots showing the probability density distribution of the calculated effect size (x-axis) of the individual studies for each test measure. The overall estimated effect size for each test measure, is indicated by a diamond, and the related 95 % confidence interval by a horizontal line. Points indicate the estimated mean effect size for each compound. Colours indicate anxiolytic compounds. Opacity is applied to not significant effect sizes, i.e. the lower bound of the 95 % confidence interval is lower than zero. An interactive version of the figure can be found online at

https://mrossovetsuisse.shinyapps.io/Shiny_SR/

Rosso M, Wirz R, Loretan AV, Sutter NA, Pereira da Cunha CT, Jaric I, Wurbel H, Voelkl B (2022) "Reliability of common mouse behavioural tests of anxiety: A systematic review and meta-analysis on the effects of anxiolytics." *Neuroscience & Biobehavioral Reviews* DOI

Treatments of Multiple Sclerosis

Systematic review and meta-analysis of 497 preclinical studies evaluating whether animal models of multiple sclerosis predict successful translation of disease-modifying therapies to clinical practice.

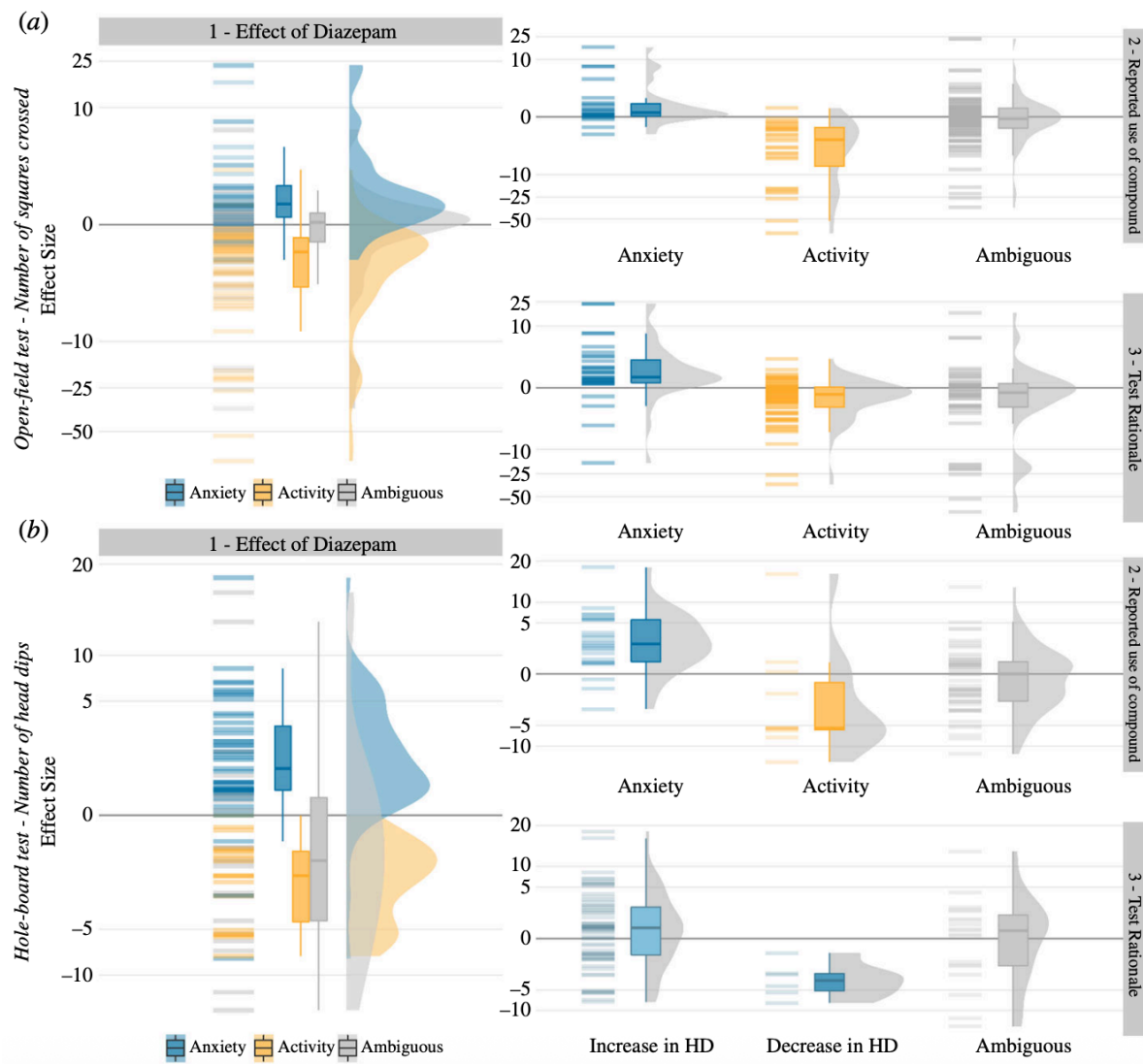


Timeline of disease-modifying therapies (DMTs) for multiple sclerosis. Approved DMTs are presented in the top panel, failed DMTs in the bottom panel. Grey circles represent individual animal studies, green circles represent first-in-MS trials and orange circles represent FDA-approval. Of note, rituximab has not obtained official FDA-approval.³¹ Abbreviations: FDA, Food and Drug Administration.

Berg I, Harvelid P, Zurrer WE, Rosso M, Reich DS, Ineichen BV (2024) "Which experimental factors govern successful animal-to-human translation in multiple sclerosis drug development? A systematic review and meta-analysis." eBioMedicine [DOI](#)

Evidence for HARKing and questionable research practices

Systematic review investigating evidence of HARKing in preclinical anxiety research by analysing how interpretations of diazepam behavioural tests varied according to study results.



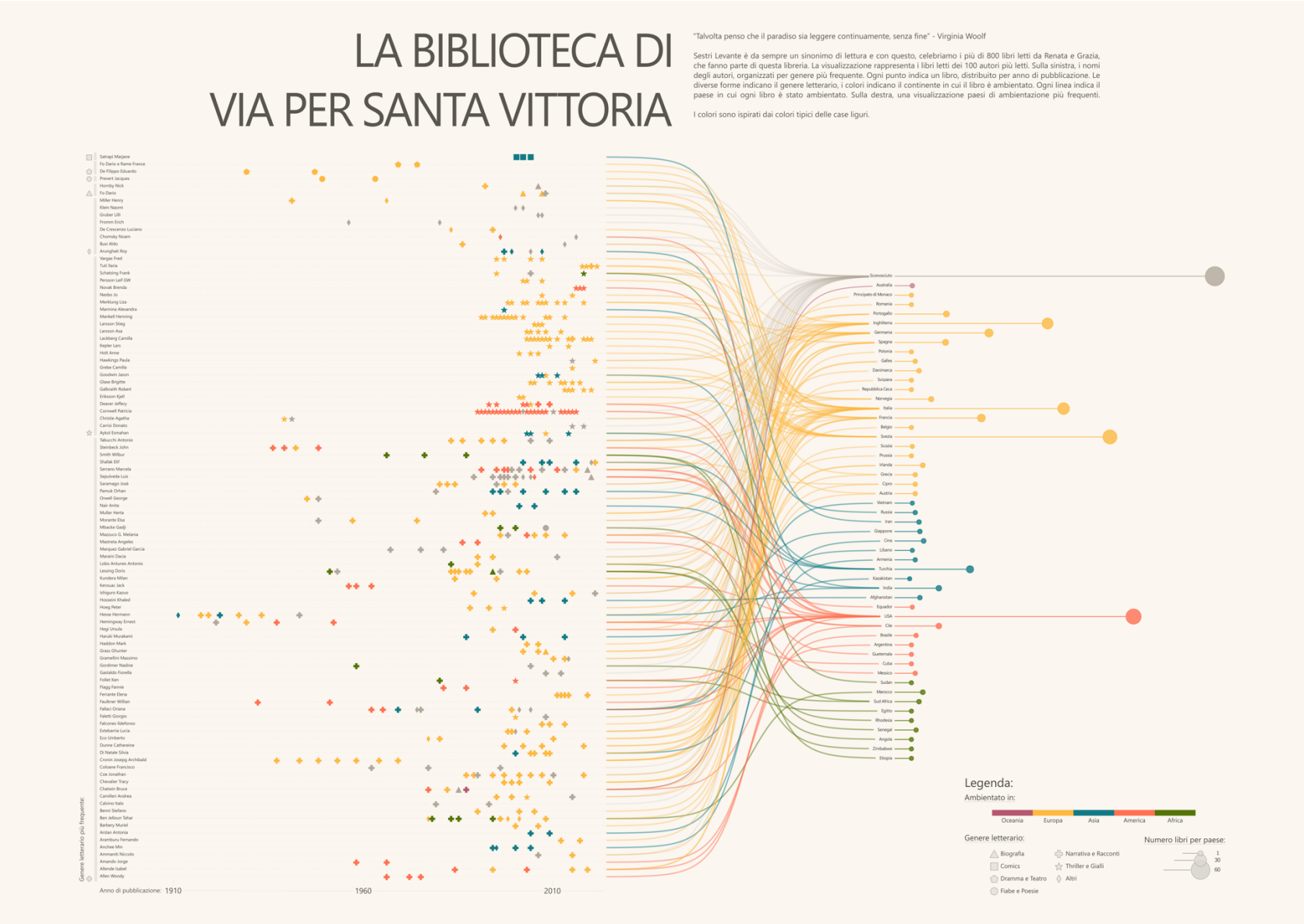
Relation between (1) effect of diazepam, (2) reported use of diazepam and (3) rationale for the use of the test measure and effect size, for the number of squares crossed in the open-field test (a) and number of head dips (HD) in the hole-board test (b). Horizontal bars, boxplots and violin plots indicate the distribution of the effect sizes as extracted from the publications.

Rosso M, Herrera A, Wurbel H, Voelkl B (2024) "Evidence of HARKing in mouse behavioural tests of anxiety." Royal Society Open Science [DOI](#)

Infographics

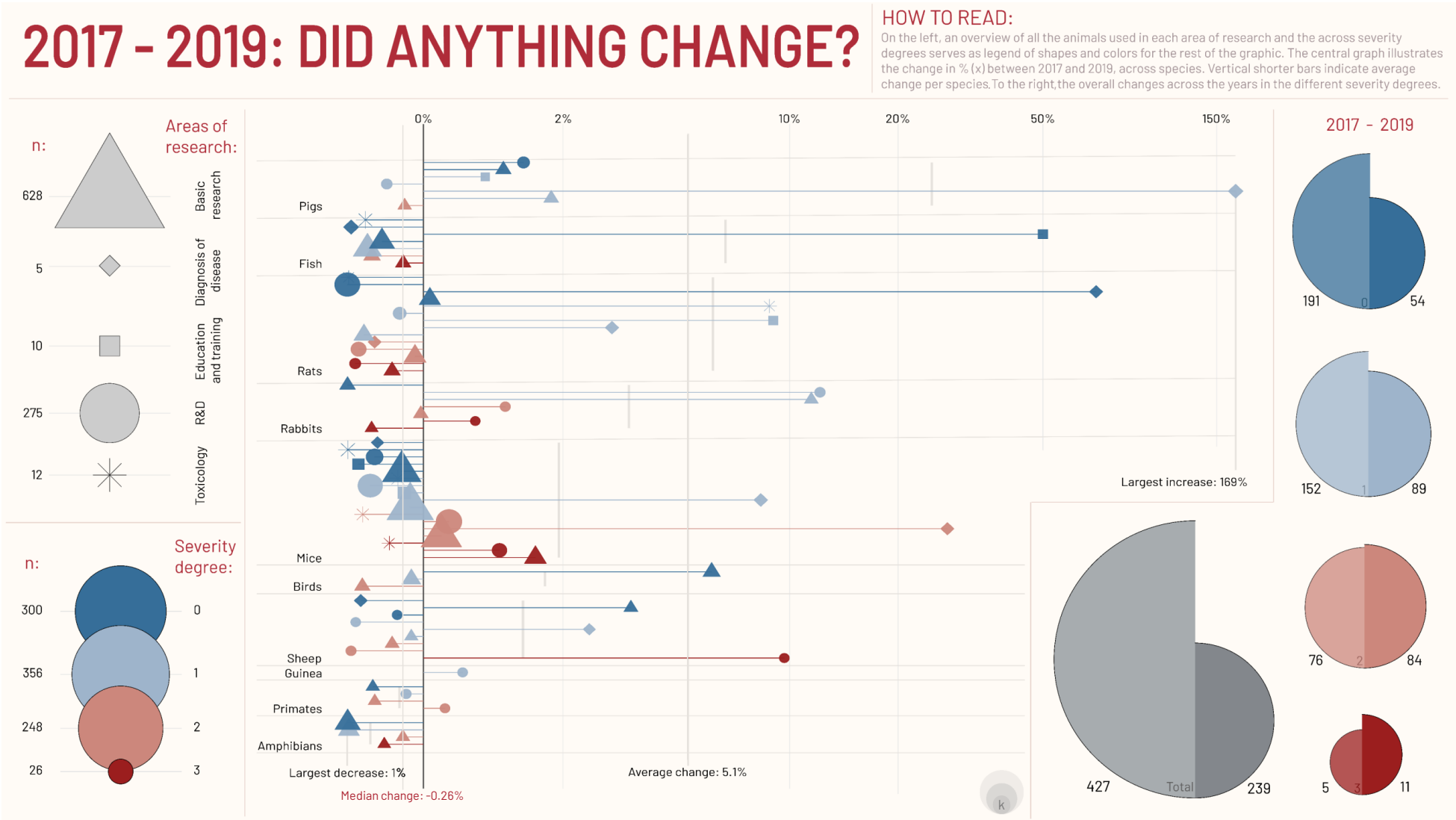
La Biblioteca di Via Per Santa Vittoria

Data visualization of a personal library integrating publication year, literary genre, author nationality, and book setting into a single narrative visualization, revealing temporal and geographical patterns across more than 800 books.



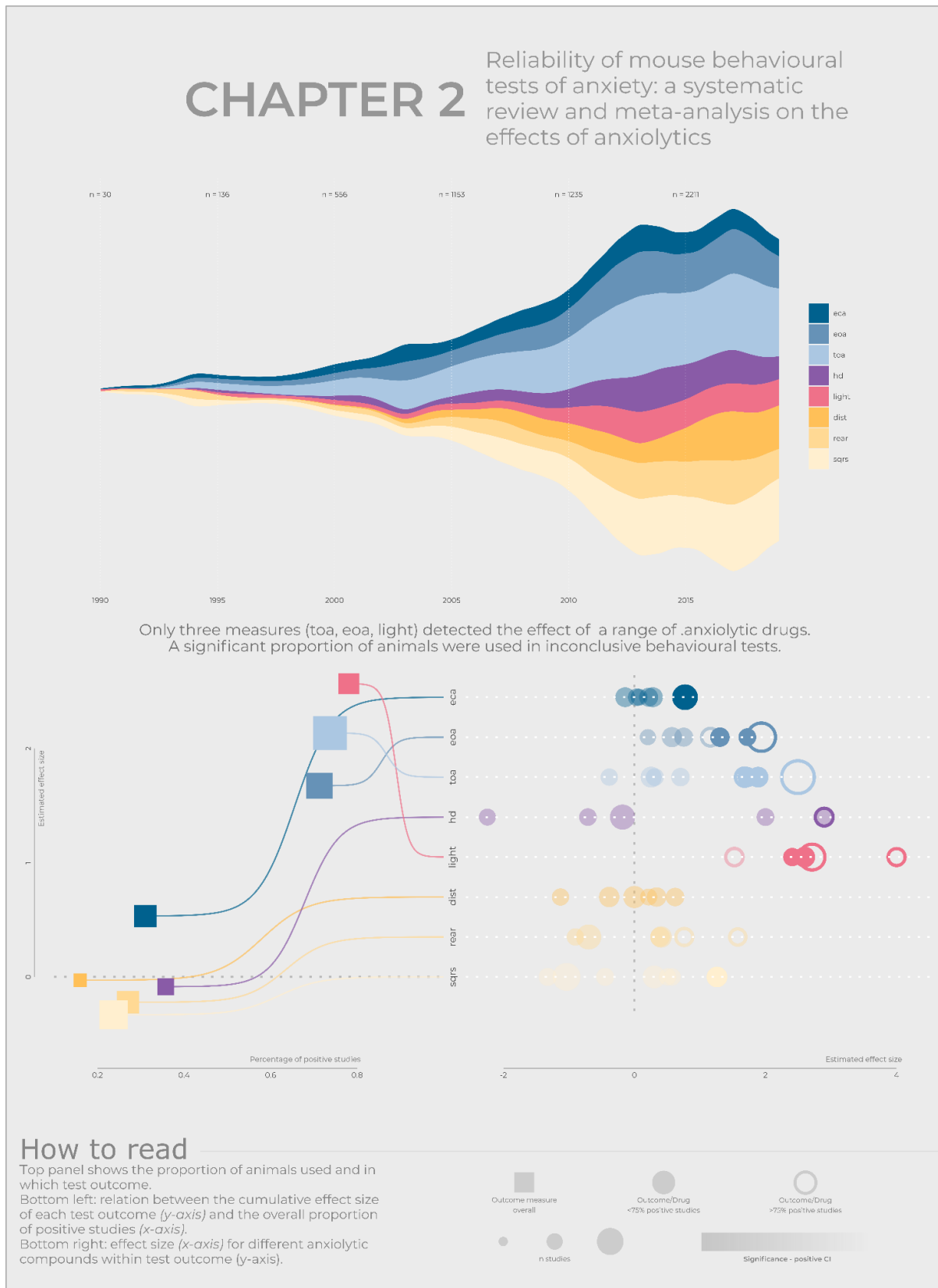
Did anything change in animal use?

Analysis of changes in animal use across research areas, species, and severity categories between 2017 and 2019, using national animal experimentation statistics.



Effects of Anxiolytics

A multi-panel infographic integrating evidence to communicate the prevalence, effect sizes, and predictive performance of behavioural outcome measures, highlighting substantial heterogeneity and the limited reliability of commonly used anxiety tests.



CLEVER-MS

Authors: Marianna Rosso, Ingrid Berg, Malcolm R. Macleod, Benjamin V. Ineichen

WELCOME

TRIALS

DRUGS

IMPRESSUM

UZH

Center for Reproducible Science



Universität Zürich



C·A·M·A·R·A·D·E·S

Review of Animal Data from Experimental Studies

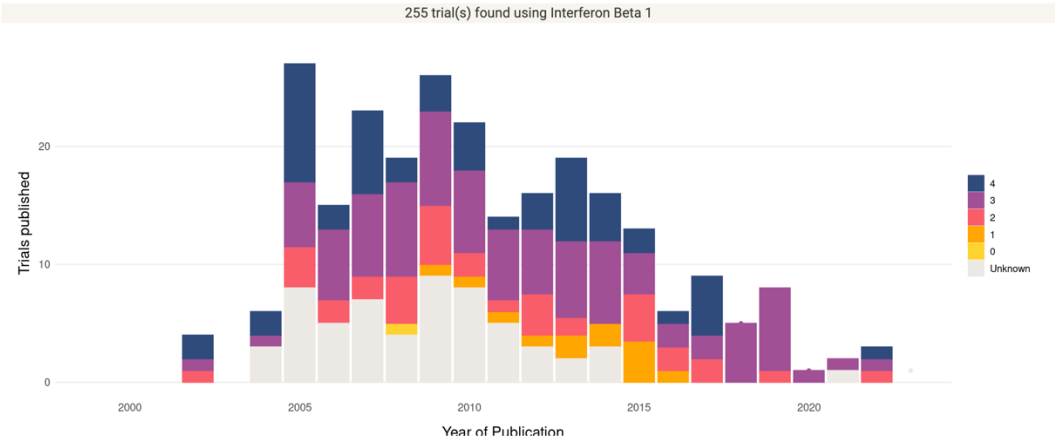
Select one treatment:

Interferon Beta 1

Select the year range:

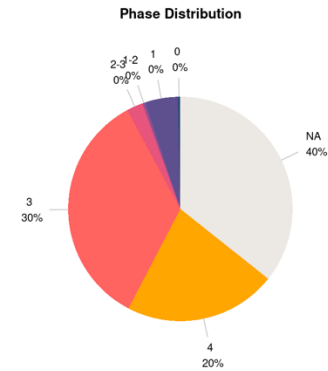
19992023

255 trial(s) found using Interferon Beta 1

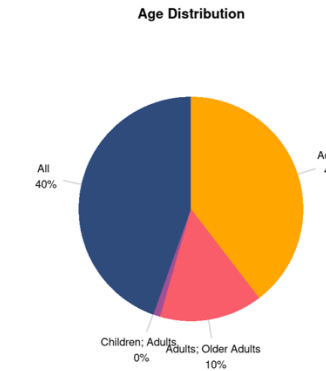


Class:	Mechanism of action:	Orphan drug status:	Orphan drug status MS:	Highest development phase MS:	Highest development phase non-MS:
Antiasthmatics; Antineoplastics; Antivirals; Interferons; Vascular disorder therapies	Immunostimulants	No	No	Marketed: MS	NA

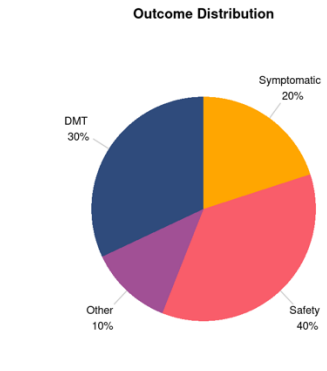
Phase Distribution



Age Distribution



Outcome Distribution



Outcomes investigated for the specific compound

33.14 %

26.18 %

14.9 %

11.44 %

4.48 %

4.38 %

2.25 %

1.31 %

1.18 %

0.56 %

0.2 %

Safety, Tolerability

Relapses

Other

Fatigue

Memory, cognition

MRI

Pharmacokinetics and -dynamics

Depression

Optical coherence tomography

Remyelination

EDSS

Mouse Anxiety Behavioural Tests Dashboard — Dashboard built from a systematic review and meta-analysis on the validity of mouse behavioural tests. (live link: [here](#))

Tool: R Shiny

Reliability of Mouse Behavioural Tests of Anxiety: a Systematic Review and Meta-Analysis on the Effects of Anxiolytics.



Tool: Tableau

